



Northern University of Business and Technology, Khulna

Final Project Report

Course Code: CSE 2216

Project Title: Dungeon Quest RPG Game

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DUNGEON QUEST RPG

Final Lab Report

1. Introduction

Dungeon Quest RPG is a simple 2D role-playing game developed using Java. The main idea of this project is to create a small adventure game where the player can explore a dungeon, fight enemies, collect experience, improve the character, and finally defeat a powerful boss.

The project was developed step by step over ten weeks. At the beginning, the main game idea and requirements were planned. After that, the basic game structure, player system, enemies, combat system, weapons, boss battle, sound effects, and other features were added gradually.

The main purpose of this project was not only to make a playable game but also to understand how different parts of a Java project work together. It helped us practice object-oriented programming, game logic, event handling, testing, debugging, and project organization.

2. Project Objectives

The main objectives of the Dungeon Quest RPG project were:

- To develop a simple and playable RPG game using Java.
- To understand how a game can be developed step by step.
- To implement player movement and interaction.
- To create enemies and a basic combat system.
- To add health, attack and upgrade systems.
- To create a boss battle.
- To design a simple dungeon environment.
- To add sound effects and improve the game experience.
- To practice Java classes and object-oriented programming.
- To test and debug the game before the final submission.
- To make the project available through GitHub so that others can view and use the project.

3. Tools and Technologies Used

The following tools and technologies were used to develop the game:

- **Programming Language:** Java
- **Graphics and Interface:** Java Swing and AWT

- **Code Editor:** Visual Studio Code
- **Version Control:** Git
- **Project Sharing:** GitHub
- **Operating System:** Windows

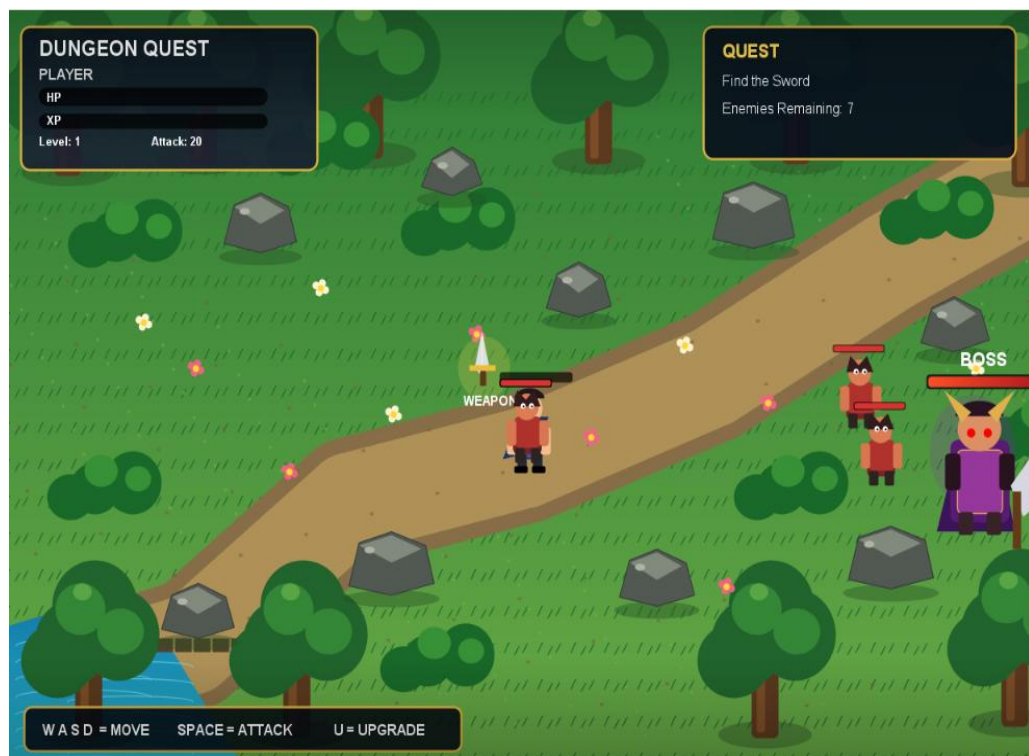
Java was mainly used for the game logic, player system, enemy system, combat and other features. Swing and AWT were used to create the game window and draw the game objects.

4. Game Overview

Dungeon Quest RPG is based on a simple dungeon adventure. The player enters a dungeon and has to explore the area while fighting different enemies. During the journey, the player can gain XP, manage health, use weapons, and improve the character.

The main goal of the game is to survive the dungeon and defeat the final boss. The game contains a simple environment with a grass field, rocks, enemies, a weapon, and other objects. The player can move around the game area using the keyboard.

The game was designed to be simple so that the main gameplay mechanics could be understood easily. Even though it is a small project, it includes several important parts of an RPG game.



5. Main Features of the Game

The main features of Dungeon Quest RPG are:

- Player movement
- Enemy characters
- Basic combat system
- Player health system
- Enemy health system
- Boss battle
- Boss HP bar
- XP system
- Weapon system
- Upgrade system
- Inventory system
- Quest system
- Dungeon environment
- Sound effects
- Game Over screen
- Game Complete screen

The player can move using the WASD keys and attack enemies using the SPACE key. The U key is used for the upgrade feature.

6. System and Class Structure

The project was divided into different Java classes. Each class was given a specific job so that the code could be easier to manage.

Main.java

The Main class is used to start the game. It creates the main game window and starts the application.

GameFrame.java

GameFrame manages the main game window. It sets the window size, title, and other basic settings.

GamePanel.java

GamePanel is one of the main parts of the game. It handles the game screen, drawing, keyboard input, and different gameplay activities.

Player.java

The Player class manages the player character. It contains player movement, health, attack, XP, and upgrade-related functions.

Enemy.java

The Enemy class is used to create and manage enemy characters. It handles enemy position, health, and interaction with the player.

Boss.java

The Boss class is used for the final boss. The boss has more health and provides a bigger challenge for the player.

Dungeon.java

The Dungeon class is related to the game environment and dungeon area.

Quest.java

The Quest class is used to manage missions and objectives in the game.

AudioManager.java

The AudioManager class is used to handle the sound effects and audio-related parts of the game.

7. Gameplay Controls

Key	Function
W	Move Up
A	Move Left
S	Move Down
D	Move Right
SPACE	Attack
U	Upgrade

These controls were kept simple so that the player can easily understand and play the game.

8. Week-by-Week Development Summary

Week 1: Project Research & Requirement Analysis

In Week 1, the basic idea of Dungeon Quest RPG was finalized. The main storyline, dungeon world, player role, missions, and game objective were planned. The required features and gameplay mechanics were also identified.

Research was done on similar RPG games to understand how character systems, combat, levels, and game design usually work. This helped us decide which features would be suitable for our project. At the end of the week, the basic requirement analysis for the game was prepared.

Deliverable: Requirement Analysis Document

Week 2: Game Design & Planning

In Week 2, the game design was planned in more detail. The player character, enemies, abilities, and their roles in the game were decided.

The dungeon environment, obstacles, background, and general game area were also planned. The complete game flow was discussed, starting from the beginning of the game and ending with the final boss battle.

Deliverable: Game Design Document

Week 3: Environment & Basic System Development

In Week 3, the Java development environment was prepared and the initial project structure was created. The main classes and basic game window were started.

The basic interface and game screen were also developed. A simple dungeon-like environment was created with a grass field and objects such as rocks. At this stage, the main purpose was to make a basic framework where the other game features could be added later.

Deliverable: Basic Game Framework

Week 4: Player Character Development

In Week 4, the player character was developed. Keyboard controls were added so that the player could move around the game area using the WASD keys.

The basic health and attack system was also introduced. The player could interact with the game environment and perform basic actions. This week was important because the player became an active part of the game instead of being only a visual object.

Deliverable: Functional Player System

Week 5: Enemy & Combat System Development

In Week 5, enemy characters were added to the game. The enemy system was created so that enemies could interact with the player during gameplay.

The basic combat system was also implemented. The player could attack enemies using the SPACE key, and the enemies could cause damage to the player. This made the game more interactive and started the main battle part of the RPG.

Deliverable: Basic Combat System

Week 6: Weapon, Item & Level System

In Week 6, the weapon and item-related features were developed. A weapon was added to the game, and the player could use the available game mechanics to improve the character.

The inventory system was also planned for managing collected items. The XP system was added so that the player could gain experience by defeating enemies. An upgrade feature was also added to improve the player's abilities.

For example, defeating a normal enemy gives XP, while defeating the boss gives a higher amount of XP.

Deliverable: Complete RPG Mechanics

Week 7: Advanced Feature Development

In Week 7, more advanced features were added to make the game feel more complete. The boss battle system was developed and a powerful boss was added.

The boss has a larger health level and provides a greater challenge than normal enemies. Quest-related features were also included. Sound effects were added to improve the gameplay experience.

The game environment and visual presentation were also improved during this stage.

Deliverable: Advanced Game Version

Week 8: Testing & Debugging

In Week 8, the game was tested carefully. Player movement, combat, health, XP, enemies, boss battle, and other features were checked.

During testing, different problems were found and fixed. Some issues were related to game logic, movement, interaction, and screen updates. The code was checked several times to make sure the game was working properly.

Deliverable: Beta Version

Week 9: Final Improvement & Documentation

In Week 9, the game was improved based on the results of testing. The game environment, interface, sound, and overall gameplay were checked again.

The source code was also organized so that different classes had their own responsibilities. Unnecessary problems and errors were fixed as much as possible.

The project documentation was also prepared during this stage. Screenshots, game information, features, and other important project details were collected for the final report.

Deliverable: Final Documentation

Week 10: Final Deployment & Live Gameplay Showcase

In Week 10, the complete game was tested again before final submission. All major features were checked together to make sure the game could be played from beginning to end.

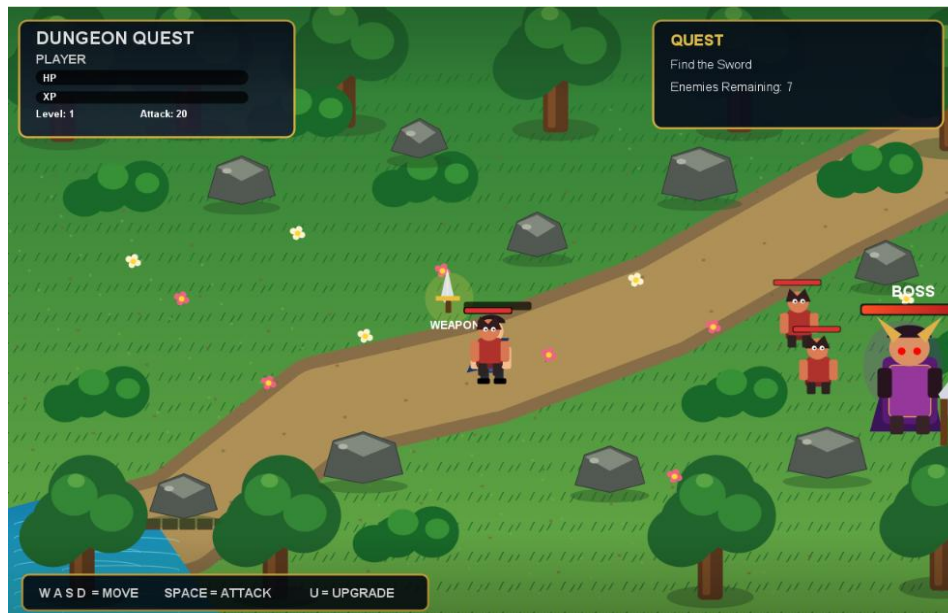
The final playable version of Dungeon Quest RPG was prepared. The project was also organized so that it could be shared through GitHub.

Deliverable: Final Dungeon Quest RPG Game

9. Game Screenshots

The following screenshots show different parts of the final Dungeon Quest RPG game.

Figure 1: Main Game Screen



The main game screen shows the player and the dungeon environment.

Figure 2: Player and Game Environment

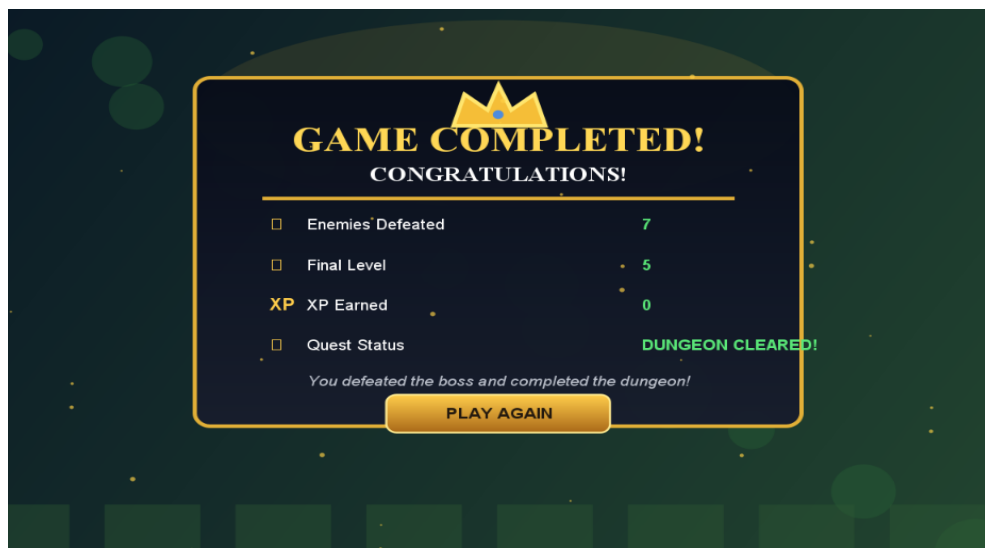


Figure 3: Enemy Combat



This screenshot shows the player fighting an enemy using the basic combat system.

Figure 4: Game Complete / Game Over



This screenshot shows the final result after completing or losing the game.

10. Testing and Debugging

Testing was done throughout the development process. Each major feature was tested after implementation. The final version was also tested as a complete game.

Test Item	Expected Result	Status
Player Movement	Player moves using WASD	Passed
Player Attack	Player attacks using SPACE	Passed
Enemy System	Enemies appear and interact with player	Passed
Health System	Health changes during combat	Passed
Upgrade System	Player can improve the character	Passed
Boss Battle	Player can fight the final boss	Passed
Boss HP	Boss health decreases during attack	Passed
Sound Effects	Game sounds work properly	Passed

Game Complete	Completion screen appears after success	Passed
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Testing helped us find small problems and fix them before the final version was prepared.

11. Problems Faced and Solutions

During the development of the game, we faced different problems. Some problems were related to Java code, while others appeared during testing.

One common problem was related to player movement and game screen updates. Sometimes the player did not behave exactly as expected. The code was checked and the movement logic was corrected.

Another problem was managing different game objects such as players, enemies, and the boss. Keeping these objects organized was important because each one had different properties and functions.

The boss system also needed extra testing because the boss had a different health level and needed to work differently from normal enemies.

Adding sound effects and making sure they worked with the game was another part that needed testing.

Through these problems, we learned that finding and fixing errors is a normal part of programming.

12. GitHub Project and How Others Can Use It

The Dungeon Quest RPG project can be shared through GitHub so that other people can see the source code, download the project, and run it on their own computer.

12.1 Requirements

To run the project, a user should have:

- A computer with Windows or a similar operating system
- Java Development Kit (JDK)
- A Java-supported code editor or IDE
- Git, if they want to clone the project
- The Dungeon Quest RPG source code

12.2 Getting the Project

A user can get the project from the GitHub repository by either downloading the project as a ZIP file or cloning the repository using Git.

After downloading the project, the user needs to extract the files and open the project folder in a suitable Java editor.

12.3 Running the Project

After opening the project, the user needs to make sure that Java is properly installed and configured.

The main Java file should then be run to start the game. The exact starting file is the `Main.java` class.

After running the program, the game window will appear and the player can start playing.

12.4 Game Controls

The controls are simple:

- W – Move Up
- A – Move Left
- S – Move Down
- D – Move Right
- SPACE – Attack
- U – Upgrade

12.5 Understanding the Project

A new developer can understand the project by checking the classes one by one.

`Player.java` contains the player-related features.

`Enemy.java` contains the normal enemy-related features.

`Boss.java` contains the boss-related features.

`GamePanel.java` manages much of the main gameplay and drawing.

`Dungeon.java` is related to the dungeon environment.

`Quest.java` handles quest-related features.

`AudioManager.java` manages the sound effects.

This class-based structure makes it easier for another developer to understand the project and make changes.

12.6 How to Add New Features

If another developer wants to improve the game, they can add new enemies, weapons, quests, levels, or other features by modifying the related classes.

For example, a new enemy can be added by working with the enemy-related code. A new weapon can be connected with the player and inventory system. New quests can be added through the quest system.

In this way, the project can be used as a starting point for making a larger RPG game.

13. Final Result

After completing all ten weeks of development, Dungeon Quest RPG became a playable Java-based RPG game.

The final version includes player movement, enemies, combat, health, XP, weapon and upgrade features, a boss battle, sound effects, and different game-ending screens.

The project also helped us understand how a complete software project can be developed step by step. Starting from a simple idea, we gradually added different features and tested them until the final version was ready.

14. Limitations

Although the game is playable, it still has some limitations.

The game has a limited number of enemy types. The graphics are simple because the main focus of the project was on learning Java and implementing game logic.

The game also does not currently have advanced features such as online multiplayer, a large number of maps, complex enemy AI, or a large story system.

These limitations can be improved in future versions.

15. Future Improvements

There are many features that can be added to make Dungeon Quest RPG better in the future.

Some possible improvements are:

- Adding more dungeon levels
- Adding more types of enemies
- Adding more bosses
- Adding more weapons
- Adding better player animations
- Adding more quests and missions
- Adding a save and load system
- Adding more advanced enemy AI
- Adding more detailed graphics
- Adding more sound and background music
- Adding different playable characters
- Adding multiplayer features

These improvements would make the game larger and more interesting.

16. What I Learned from This Project

Working on Dungeon Quest RPG was a very useful experience for me. Before starting the project, I mostly understood programming from the theory and small coding problems. While making this game, I got the chance to use those ideas in a real project.

I learned how different Java classes can work together to create one complete application. I also learned how to handle player movement, enemies, health, attacks, XP, upgrades, and other game features. It helped me understand that every small feature needs proper planning before adding it to the project.

One of the biggest things I learned was debugging. During the development, the game did not always work as I expected. Sometimes a small change caused another problem. I had to check the code, find the reason, and test it again. This made me more patient and careful while programming.

I also learned that testing is not something that should only be done at the end. Testing each feature after adding it makes it much easier to find problems.

Another thing I learned was how important it is to keep the project organized. Using separate classes for the player, enemies, boss, quests, dungeon, and audio made the project easier to understand and change.

Working with GitHub also gave me an idea about how a project can be stored and shared with other people. Someone else can get the source code and continue working on it.

Overall, this project gave me practical experience in Java and helped me become more confident in programming. More importantly, I learned that building a complete project takes time, testing, mistakes, and continuous improvement. Making Dungeon Quest RPG from the first idea to a playable final game was a valuable learning experience for me.

For my next game project, I will focus more on these areas:

1. **Proper Planning**
Before coding, I will plan the story, characters, levels, enemies, missions, controls, and main goals of the game.
2. **Clear Game Concept**
I will make sure the main idea of the game is clear, including what the player needs to do, how they can win, and how they can lose.
3. **Step-by-Step Development**
Instead of building everything at once, I will develop the game in small parts, such as movement, attack, enemy, health, levels, and boss battles.

4. **Clean and Organized Code**
I will keep the code more organized by using proper classes and methods. This will make it easier to fix problems and add new features later.
5. **Regular Testing**
I will test each feature after developing it instead of waiting until the whole game is finished.
6. **Game Design and User Experience**
I will focus more on making the controls easy, the interface clear, and the overall gameplay enjoyable for the player.
7. **Better Graphics and Sound**
I will try to improve the graphics, animations, background, UI, and sound effects to make the game more attractive.
8. **Balanced Difficulty**
I will make sure the game is not too easy or too difficult. The difficulty should increase slowly as the player moves forward.
9. **Save and Progress System**
If possible, I will add a system to save the player's progress, level, score, and inventory.
10. **Easy Future Updates**
I will design the code in a way that makes it easier to add new characters, weapons, enemies, levels, and missions later.